

TUTORIAL-1

1. For the circuit in Fig.1a the current is known to be zero at $t = 0$. The voltage across the 3-H inductor is given in Fig.1b.
 - (a) Plot the current as a function of time.
 - (b) The 0.5-F capacitor is initially discharged. Plot the voltage $v_C(t)$ as a function of time across the capacitor.

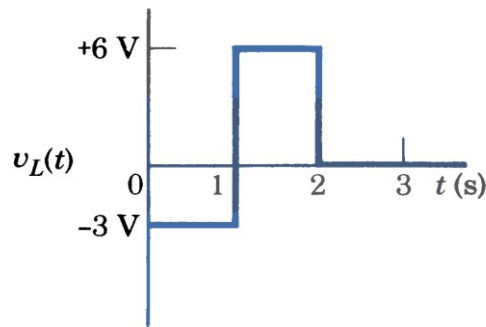
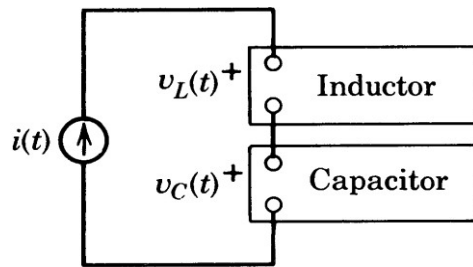
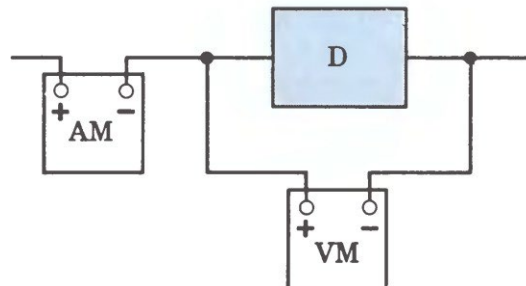


Fig. 1

2. Ammeter AM and voltmeter VM, connected as shown in Fig.2, measure instantaneous current and voltage. An ammeter reads “up- scale” or positive when current flows into the meter at the + terminal. Is power flowing into or out of device D when:

- (a) AM reads positive and VM reads negative?
- (b) AM reads negative and VM reads positive?
- (c) AM reads negative and VM reads negative?



3. The voltage source in circuit of Fig 3.0 is $v = 200t$ V, where t is in seconds,
- For $t = 0.5$ s
 - Calculate the power dissipated in the $200\text{-}\Omega$ resistor.
 - Calculate the power absorbed in the $1000\text{-}\mu\text{F}$ capacitor.
 - Power delivered by the source.
 - Total energy stored in the capacitor.
 - Now assume $v = 2000t$ V, where t is in seconds. For $t = 5$ ms, find the current through the $4\text{-}\Omega$ resistor and through $300\text{-}\mu\text{F}$ capacitor.

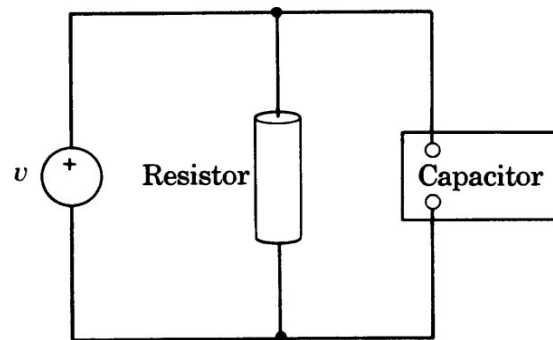


Fig. 3

4. Devise a linear model incorporating a current source to represent the solar battery whose output characteristics are shown as in Fig.4. Specify numerical values valid for currents up to 40 mA.

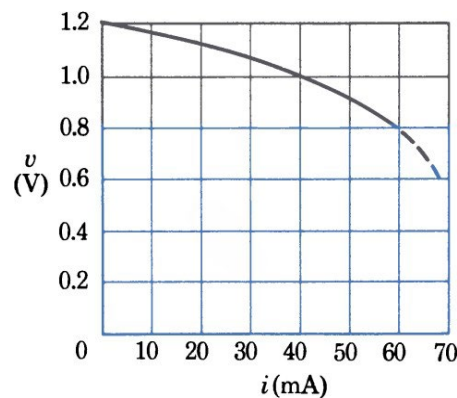


Fig. 4 Solar battery characteristic

5. Each of the following mathematical models describes either a resistance, capacitance, or inductance. Show which one and why for each model.
- $v = kq$
 - $v = k \frac{dq}{dt}$
 - $v = k \frac{d^2q}{dt^2}$
6. The circuit shown in Fig. 5 can be used to smooth out or filter variations in input current i so that a reasonably constant voltage appears across load resistor R . For good filtering, the capacitor should be able to store ten times as much energy as is dissipated in one cycle by R . If $R = 10\text{K}\Omega$ and variations in i occur 60 times per s, specify the value of C

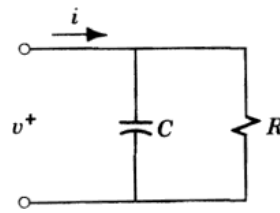


Fig. 5

7. Laboratory tests on a coil yield the following data: a change in current from +10 mA to -10 mA in 50 μ s results in an average voltage of 2V; a steady current of 10mA requires a power input of 0.2 W. Devise a linear circuit model to represent the coil.
8. A fuse protects a circuit by melting when the current becomes excessive. The resistance of the fuse is given by $R = R_r(1 + cT)$ where R_r is the resistance at room temperature, c is the temperature coefficient, and T is the rise above room temperature. The rise in temperature is proportional to the heat generated in the fuse or $T = kP$. Derive an expression for R in terms of current I and evaluate the current at which the fuse “blows”.
9. A rechargeable flashlight battery is capable of delivering 90 mA for about 12 hours. How much charge can it release at that rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?
10. The current through an element is shown in the figure below. Determine the total charge that passed through the element at:
 - (a) $t=1$ s
 - (b) $t=3$ s
 - (c) $t=5$ s

